



Unit 4 · Outcome 3 · Scientific Investigation

Assumptions & limitations

KK09

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Assumptions — the things you take as true

01 Assumptions — the things you take as true A premise your method or conclusion quietly depends on. An assumption is a condition you accept as true without directly proving it, so that your investigation makes sense and your conclusion holds. Assumptions are usually unstated — which is exactly why examiners reward students who can drag them into the light. ♦ Analogy · the foundations you never see A conclusion is a house. Assumptions are the foundations underneath it — invisible once the house is built, but the house only stands if they hold. If a key

Limitations — what your study could not do

02 Limitations — what your study could not do A factor that restricts the investigation, or narrows the conclusion you can fairly draw. A limitation is a feature of your design, equipment or scope that restricts how confidently or how widely you can conclude. Crucially, a limitation is built in — it would still be there even if you ran the whole thing perfectly, with no slips. ♦ Analogy · the size of your window A limitation is the size of the window you're looking through. Dev studied one roof, one tree, one fortnight — a small window. Whatever Dev se

The three sources VCAA names

03 The three sources VCAA names The dot point lists them explicitly: methodology · data generation · analysis. The wording of KK09 is a gift — it tells you exactly where to go looking. Strong students walk through all three categories like a checklist, which is also how the marks are usually split. Methodology & method How the study was designed and run. Small sample / few replicates Narrow range of the independent variable Short time frame; single site or season A confounding variable you couldn't control No genuine control group Data generation How t

Assumption vs limitation vs error

04 Assumption vs limitation vs error The distinction examiners say students get wrong most often. Three quick tests will place almost any statement: Assumption — is it a premise you're taking as true for the study to work? "We assume X is true." Limitation — would it still be there if you ran the study flawlessly? Then it's built into the design or scope. "The study can't..." Personal error / mistake — is it an avoidable slip a careful person could've prevented? Then it's an error, not a limitation. "I accidentally..." Sort the statements drag · or tap-then-

From limitation to improvement

05 From limitation to improvement A limitation is only half a mark; the improvement that fixes it is the other half. The poster's discussion — and most exam "evaluate" questions — want a limitation paired with a realistic improvement that actually addresses it. The trap is offering "be more careful," which fixes error, not the limitation. A real improvement changes the design, equipment or scope. Match the fix pair Tap a limitation on the left, then tap the improvement that genuinely addresses it on the right. Limitation Improvement that fixes it 0 /

Writing it the way examiners reward

06 Writing it the way examiners reward Decode the command word, then build with P·E·L. First, read the verb. The same scenario earns very different things depending on the command word: Identify / State Describe Explain Evaluate Justify / Suggest Build the answer with P·E·L For a limitations response, Point (name it + its category), Example (the specific detail from the study), Link (what it does to the conclusion + the fix). Type below and the meters track whether you've hit each move. P·E·L writing trainer live feedback Prompt: A student measures strea

Six exam traps

⚠ EXAM TRAPS

!! Six exam traps Where U·O3 marks quietly leak away. ⚠ Trap 1 · calling a mistake a limitation "I read the scale wrong" / "I forgot to start the timer" are personal errors . They're avoidable and don't belong in a limitations answer. A limitation survives a perfectly-run study. ⚠ Trap 2 · "human error" as a catch-all Writing "human error" earns nothing — it's vague and usually means a mistake anyway. Name a specific, structural limitation instead. ⚠ Trap 3 · a limitation with no consequence "Small sample size" on its own is a list item, not an answer. S

Practice exam · 20 marks

★ PRACTICE & SELF-MARK

EX Practice exam · 20 marks VCAA-style. Reveal weak / strong / marking guide, then self-mark. Your score tracks in the dock. Q1 1 mark Multiple choice A student investigating earthworm density assumes that "the soil moisture was the same across all five plots." This statement is best classified as: A. a personal error B. an assumption C. a methodology limitation D. a random error Weak answer Strong answer Marking guide C — methodology limitation. Why it drops the mark: it confuses a stated premise taken as true with a constraint on the design. The studen

One-screen cheat sheet

✦ RECAP / CHEAT SHEET

TL One-screen cheat sheet Everything KK09, compressed for the night before. // definitions Assumption — premise taken as true; the conclusion depends on it. Limitation — built-in constraint; restricts the conclusion. Error — avoidable slip; not a limitation. // three sources Methodology — sample size, range, time, site, confounders, no control. Data generation — resolution, subjectivity, too few repeats. Analysis — correlation ≠ cause, no extrapolation, small n. // the three tests "We assume X." → assumption Survives a perfect run? → limitation A careful

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

C — methodology limitation. Why it drops the mark: it confuses a stated premise taken as true with a constraint on the design. The student isn't saying the study couldn't control moisture — they're assuming it was equal.

✓ STRONG / MODEL ANSWER

B — an assumption. Why it's right: the word "assumes" and the structure "X was the same" mark it as a premise accepted as true so the comparison is fair — the definition of an assumption.

✗ WEAK ANSWER

An assumption is something you think is true and a limitation is a problem with the experiment." Why it caps at 0–1: "a problem with the experiment" is too loose — it doesn't separate a constraint from a mistake, and "something you think is true" misses that the study depends on it.

✓ STRONG / MODEL ANSWER

An assumption is a premise accepted as true without proof, that the method or conclusion relies on. A limitation is a feature of the design, equipment or scope that restricts how confidently or how widely the conclusion can be drawn — and would remain even if the study were run perfectly." Why it's full marks: defines both precisely and draws the line — depends-on vs restricts — without slipping into "error."

✗ WEAK ANSWER

Dev might have read the app wrong, and the experiment wasn't very long so it could have human error." Why it scores ~0–1: "read the app wrong" and "human error" are avoidable mistakes, not limitations. "Wasn't very long" gestures at a real point but has no category and no consequence.

✓ STRONG / MODEL ANSWER

Methodology: the study ran on one roof with one tree across a single two-week April window, so the sample is narrow and seasonal — the sun is lower in April than summer, so the result may not hold across the year. Data generation: the inverter logged only daily totals to the nearest 0.1 kWh, so small differences between arrays may fall below the resolution and within-day shading effects are invisible. Why it's full marks: two limitations, two different categories, each with a specific detail and a stated consequence.

✗ WEAK ANSWER

Dev assumed the panels worked. A limitation is that it could be inaccurate, so Dev should be more careful and do it again." Why it scores ~1/5: "panels worked" is trivial and unexplained ; "could be inaccurate" is vague; and "be more careful / do it again" fixes error, not a limitation — no design change.

✓ STRONG / MODEL ANSWER

Assumption: Dev assumed the two arrays were identical apart from the shading . This matters because any built-in difference — age, tilt, wiring — would otherwise explain the output gap instead of the tree, undermining the comparison. Limitation & effect: Dev could not control daily cloud cover (a confounding methodology limitation). On cloudier days both arrays drop, so day-to-day differences may reflect weather, not shade — weakening the validity of any single-day comparison and the strength of the conclusion. Improvement: mount a reference cell / pyranometer to measure incoming irradiance, then compare output relative to available sunlight — so cloud is accounted for rather than left as a

✗ WEAK ANSWER

Aria assumed birds were there. The limitation is she might have miscounted or scared the birds, and she only did it once which is human error. To improve she should count more carefully and use a better method." Why it scores ~2/8: the assumption is trivial; "miscounted / scared the birds / human error" are avoidable mistakes ; "once" is a real point but uncategorised with no consequence; and the improvements are "be careful," not design changes . No category structure.

✓ STRONG / MODEL ANSWER

Assumption: Aria assumed the two verges were comparable in every way except the revegetation (soil, traffic, adjacent habitat). The conclusion that revegetation drives any difference relies on this; if the control verge differed in, say, traffic noise, that could explain the gap instead. Methodology limitation: a single 20-minute count at one site pair on one morning gives a tiny, unreplicated sample. Bird activity varies hugely with time of day, weather and season, so this snapshot is unlikely to be representative — the conclusion can't be generalised beyond that morning. Improvement: repeat counts across multiple mornings, times and several matched site-pairs. Data-generation limitation: r